Legal Framework for Open-Sourcing Language Data in India

A white paper to inform stakeholders and ecosystem

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I. Genesis and Acknowledgements

The idea and requirement for this white paper stem from a collaborative workshop that brought together 17 organisations actively engaged in the collection and dissemination of Indian language speech data. The workshop was organised by EkStep Foundation and IISc/ARTPARK. The discussion on legal framework for dataset open sourcing, facilitated by Mr. Rahul Matthan and Mr. Amlan Mohanty, highlighted the urgent need for a clear legal framework to guide the open-sourcing of such data. During the workshop, participants expressed a strong desire to make their collected data available to researchers and innovators, ideally through BHASHINI. However, they emphasized the necessity of legal clarity to ensure legal safeguards for their organisations to prevent potential liabilities associated with open-sourcing data.

As an action item from this discussion, the drafting of this white paper was identified as a critical next step. The collaborative effort of the participating organisations have been instrumental in bringing this document to fruition. While it is not feasible to credit each organisation individually as authors, their contributions and engagement are gratefully acknowledged.

The following organisations participated in the workshop and contributed to the discussion:

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This white paper aims to provide the necessary legal framework to support the open-sourcing of Indian language speech data, reflecting the collective input and vision articulated during the workshop.

II. Introduction

Given the growing demand for artificial intelligence (**AI**) models localized to the Indian experience there is an urgent need for reliable datasets. This includes speech data in regional languages, to facilitate the widespread implementation of speech language technologies among India's marginalized communities. These speech datasets, comprising of audio files, underlying text/transcripts, and annotations/metadata (**Speech Datasets**), have diverse applications, from Automated Speech Recognition to Text-to-speech systems. Many organisations in India are currently engaged in, or are planning to engage in, collecting and hosting Speech Datasets (**Organisations**) with the aim of creating a substantial open-source repository adaptable to various contexts. In connection with this, there are certain legal and ethical considerations that Organisations need to bear in mind across the lifecycle of a Speech Dataset.

This document is intended to serve as a foundational resource for Organisations seeking to understand and appreciate the legal issues pertaining to the creation and hosting of Speech Datasets in India. It is not legal advice and Organisations should seek legal counsel to advise them on specific situations and challenges.

Section II of this whitepaper provides a broad overview of the legal framework surrounding the creation of Speech Datasets, namely (i) the laws governing ownership of Speech Datasets and (ii) the regulatory liabilities associated with creating and managing Speech Datasets. Section III discusses the various legal considerations to be kept in mind by Organisations to ensure that they have sufficient rights to use Speech Datasets at all stages of the dataset lifecycle - from collection to development and curation, and finally distribution of the datasets. Section IV discusses regulatory liability considerations that would be relevant across the lifecycle of a Speech Dataset. Finally, Section V sets out certain ethical best practices to be considered by Organisations from a reputational perspective.

III. Legal Framework: Overview

Under the existing Indian regulatory regime, different legal frameworks govern ownership rights and liabilities in relation to Speech Datasets. This section provides an overview of the intellectual property (**IP**) and data protection concerns that Organisations will have to consider.

(A) Ownership: Intellectual Property Rights

Organisations need to ensure that they have sufficient rights over Speech Datasets so that they are able to curate, store, host, share and otherwise manage Speech Datasets as they see fit as well as to create an open-source repository. Under Indian law, ownership over datasets is primarily governed by various IP laws, which govern the intellectual property rights with respect to different categories of works. Rights over datasets are primarily conferred through copyright, and governed by the Copyright Act, 1957 (**Copyright Act**). As per the Copyright Act, copyright subsists in various works including literary works and sound recordings. Copyright would therefore subsist in various elements of a Speech Dataset, such as the underlying text/transcripts and metadata, as well as the voice recordings.

Copyright may further subsist in extracts of works, to the extent such extracts form an essential part of the work of the person in whom the copyright lies. However, no copyright subsists in raw data or information (i.e. data which has not been expressed or organised in a unique and original way such as a database) *per se*.

Transfer Rights

The Copyright Act grants exclusive rights to the author or original creator of a piece of work, which includes the right to reproduce, distribute, and display the work. To use or exploit such work, third parties may obtain (i) an assignment of the copyright which confers ownership of such copyright (or any part of the rights), or (ii) a license which confers a right to use the copyright (or any part of the rights, in that work from the owner in accordance with the provisions of the Copyright Act.

For full transfer of rights and to ensure complete protection, an assignment should contain the following essential elements: (i) active assignment of the intellectual property (it should contain language that explicitly states that the copyright has been assigned else the assignment may be deemed contingent); (ii) the term and territory of the assignment (in the absence of which it will be assumed that the term of assignment is limited to 5 years and the territory is limited to India); (iii) the amount of royalty and any other consideration payable to the voice artist or their legal heirs during the term of the assignment; (iv) waiver of reversionary rights (in the absence of which the assigned intellectual property will revert back to the assignor if the assignee has not used the work for a year); (v) an undertaking to not challenge the assignment or the ownership of the assigned copyright or any modifications (together, **Assignment Requirements**).

Further, under the Copyright Act, performers (including persons engaged by the Organisation to lend their voice for creating a voice recording) have certain limited rights and are entitled to an equal share in royalties in case of making of the performances being made available for commercial use.

Open Content Licenses

Copyright owners may also choose to waive their rights in the work and make it freely available for public use. Such open material is usually subject to certain globally recognised terms, which aim to enhance access to material while ensuring the end users attribute appropriate credit to the original author. In relation to content other than software, open content licenses such as the various Creative Commons license may be utilised, including Creative Commons Attribution-NonCommercial 4.0 License (**CC BY-NC 4.0**), Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License (**CC BY-NC-SA 4.0**) Creative Commons CC0 Public Domain Dedication License (**CC0 Public Domain Dedication License**), Creative Commons Attribution 4.0 International License (**CC BY 4.0**), and Creative Commons Attribution Share Alike 4.0 International License (**CC BY-SA 4.0**) (**Open Content Licenses**). Entities using Open Content Licenses must comply with the terms of the applicable license. We have provided an overview of these licenses under Annexure - II.

Section III of this document discusses in greater detail, how Organisations can ensure copyright ownership over these various elements of a Speech Dataset.

(B) Regulatory Liability: Data Protection

Certain categories of data within a dataset are subject to compliance requirements under Indian data protection laws. Organisations must ensure that they are aware of how to collect, store, manage and share these categories of data to avoid regulatory action and penalties for non-compliance. As relevant to Speech Datasets, they may contain data about individuals who are identifiable by such data or in relation to such data (**Personal Data**). Individuals may be identifiable through various elements of a Speech Dataset including (i) their voices *per se* as part of the voice recording; (ii) information contained in the underlying text/transcript of a voice recording; (iii) metadata created in relation to a voice recording; (iv) any information voluntarily provided by an individual when they are engaged by an Organisation to lend their voice for creating a voice recording; and (v) information collected from end-users of a Speech Dataset.

Data Protection Compliances

The manner in which Personal Data is collected and processed is regulated by Indian data protection law, which prescribes various compliance requirements described in Annexure I (**Data Protection Compliances**). In this regard, India has recently introduced the Digital Personal Data Protection Act, 2023 (**DPDPA**), which introduces uniform compliance for processing all types of Personal Data. Under the DPDPA, consent of the providers of information forms the primary basis for processing Personal Data. Consensual processing of Personal Data requires data principals to give their free, informed, specific, unambiguous and explicit consent to processing, and must be

preceded by a notice which provides them information about the Personal Data and the purpose for which it is processed. Other obligations of data fiduciaries for processing personal data includes implementing security standards, ensuring completeness, accuracy and consistency of personal data, certain retention restrictions, and the provision of certain rights and enabling data principals to exercise their rights, as further detailed in Annexure I. Notably, the DPDPA does not apply in respect of processing personal data for purposes necessary for research if the personal data is not used to take any decision specific to a data principal and such processing is undertaken in accordance with the prescribed standards. However, the scope of this exemption is unclear until further clarity is provided by the Indian Government.

Until the DPDPA comes into force, Personal Data will continue to be governed by the Information Technology (Reasonable Security Practices and Sensitive Personal Data or Information) Rules, 2011 (**Privacy Rules**) which governs how entities collect, store, process, transfer or disclose personal information (**PI**)[1] or sensitive personal information (**SPI**)[2]. For instance, entities are required to publish a privacy policy, provide certain access and correction rights. With respect to SPI, they must also obtain consent for collection of SPI and comply with various compliances in relation to storage, security and transfer of SPI, as further detailed in Annexure I.

Section IV of this document discusses how Organisations can ensure compliance with Indian data protection laws with respect to the various elements of a Speech Dataset in greater detail.

[1] **Note:** Privacy Rules defines PI as 'Any information that relates to a natural person, which, either directly or indirectly, in combination with other information available or likely to be available with a body corporate, is capable of identifying such person'.

[2] **Note:** Privacy Rules defines SPI to mean passwords, financial information, physical, physiological and mental health condition, medical records and history, or biometric information. Biometrics is further defined as ‘technologies that measure and analyse human body characteristics, such as 'fingerprints', 'eye retinas and irises', 'voice patterns', ‘facial patterns', 'hand measurements' and 'DNA' for authentication purposes.

IV. Ownership Rights

The creation and dissemination of a Speech Dataset involves various stakeholders, including the Organisation conceptualising and creating the dataset, its personnel, third party organisations who may be engaged to create underlying text samples or undertake recording, entities providing open-source text databases, voice actors, and users of the dataset, each of which may hold certain rights over their contribution to the dataset. Given the various entities involved across the lifecycle of a Speech Dataset, it is important that Organisations understand the legal framework governing IP ownership, to ensure that they obtain and protect the necessary rights over the Speech Datasets they create at every stage.

(A) Collection

When creating or collecting voice recordings, Organisations should account for, and obtain ownership rights over each element of a Speech Dataset, as follows:

(1) Ownership over voice/audio recording:

As per the Copyright Act, in the event that an Organisation is responsible for the creation of a voice recording and arranges for its production, it would be considered the 'producer' of a sound recording and would be the first author of such sound recording. Therefore, all copyright in such sound recording will originally vest in the Organisation as the owner. However, the voice artist may have certain 'performer's rights' in relation to the voice recording which *inter alia* allows the voice artist to make a sound or visual recording of the 'performance'. Voice artists would also have 'moral rights', which irrespective of assignment, grants them the right to be identified as the performer in certain circumstances, and to restrain or claim damage for any distortion, mutilation, or other modification prejudicial to the voice artist's reputation.

Therefore, to the extent the voice artists may have such limited IP rights in the voice recordings, Organisations must ensure that voice artists assign these rights to the Organisation by signing a consent form/agreement (**Consent Form**), containing language to cover all the Assignment Requirements. In this regard, based on the intended end uses of the Speech Datasets, it is recommended that the territorial extent of the assignment is worldwide, and the time period is made perpetual.

If the voice actors are paid a commensurate sum in lieu of the assignment of their rights, it would strengthen the validity of the assignment. In the interest of transparency and to avoid any future claims, Organisations should also explain the terms of the assignment to the voice actor in a manner that they understand, at the time of execution of the Consent Form.

Where Organisations procure voice recordings from third-parties, Organisations must require such third-parties to assign the copyright in such works. In addition to the Assignment Requirements, contracts for assignment must also include adequate representations and warranties from such third parties to ensure that they are the

rightful owners/have all necessary rights required to validly assign the voice recordings to the Organisations, and that such assignment does not violate or infringe intellectual property rights of any other person. It is also recommended that Organisations obtain corresponding indemnities in case of any infringement claims against them in relation to their use of the assigned voice recordings.

(2) Ownership over underlying text/transcription

All underlying text excerpts used for voice recordings would constitute a literary work, with all copyrights first vesting in the author of such work. Ownership over underlying text would vary based on the manner in which the Organisation obtains such text:

- Where an Organisation itself develops this text, it would have all rights over the text and would not require any third-party assignment.
- If underlying text is created or commissioned by a third-party entity or individual, the Organisation must ensure that full rights over such text is assigned to the Organisation in accordance with the Assignment Requirements on a worldwide and perpetual basis.
- Where Organisations rely on publicly available text databases made available under the Open Content Licenses, Organisations must ensure that they comply with the licensing conditions associated with the respective textual sources.

There may also be a risk that open-source text databases or publicly available sources may contain copyrighted material such as sentences which infringe on third-party copyright. To mitigate the risk of any infringement claims, it would be advisable for Organisations to adopt quality checks on the text excerpts prior to preparing the voice recordings. For instance, Organisations may appoint individuals to manually review the text excerpt to determine whether large quantities of text appear to be taken from copyright protected sources. Excerpts may be modified as required such that they contain randomized and limited quantity of data from any given source to ensure that there is no co-relation between the sentences, or such that the text refers only to facts without containing any opinion or insight.

(B) Curation and Development

(1) Ownership over metadata

Where an Organisation creates metadata in relation to the voice recordings, such metadata is likely to constitute a literary work under the Copyright Act. As the author of such work, the Organisation will have ownership rights over such metadata in the first instance and would not require any third-party assignment.

(2) Ownership of IP created by employees of the Organisation

The Organisation may engage the services of various persons – its own employees, third-party developers, human annotators, analysts, researchers, and project associates (together, **Personnel**) engaged for the purpose of conducting research,

analyzing the voice recordings, and performing various functions on the voice recordings to model it for further usage. Such Personnel may make significant contributions towards the creation of Speech Datasets. Organisations should take steps to ensure that ownership of all such contributions is validly assigned to the Organisation.

In the case of employees, the Copyright Act states that the copyright in contributions made by employees automatically vest in the Organisation. However, where individuals are engaged as contractors or consultants, they will retain ownership rights over such contributions unless validly assigned. In any case, to ensure protection and strengthen their existing ownership rights (if any), the Organisation should consider introducing Assignment Requirements in their engagement or employment letters with all Personnel, regardless of their employment status. It is recommended that such agreements also contain robust confidentiality clauses to protect any proprietary information and processes that may be used or undertaken during development.

(C) Deployment and Public Sharing

Prior to making the Speech Datasets available for public use under an open-source license, and once all relevant contributors have adequately assigned their IP rights to the Organisation, the Organisations may choose to share Speech Datasets under any Open Content License of its choice, basis the license terms that best suit its purposes. All sharing must be undertaken in accordance with the terms of the chosen Open Content License. When choosing an appropriate Open Content License, Organisations must be mindful of the following:

(1) Restrictive Provisions

If they have created the Speech Dataset using underlying text data or training datasets made available by third parties under Open Content Licenses, Organisations must ensure that they comply with the requirements of this original license while choosing a license for the Speech Datasets. For instance, it may have obtained underlying data under an Open Content License with copyleft provision such as CC BY-SA 4.0 which has a ShareAlike[3] condition requiring the Organisation to share Speech Datasets under the same terms. Further, if the original license does not permit use for commercial purposes, Organisations must ensure that the Speech Datasets are made available free of charge or at the very least, without incurring any profit from such sharing.

(2) Royalty Payments

Performers such as voice artists have an irrevocable right to claim royalty in case of commercial use of the work. However, a performer's right to receive royalty has been understood to encompass commercial use only where the use of the

[3] **Note:** Organisations remixing, transforming, or building upon the material, are required to distribute their contributions under the same license as the original.

performer's work would result in the performer "*suffering a loss of their legitimate dues*" or the user incurring illegal gains. Therefore, if Organisations intend that the open-source Speech Datasets are to be used free-of-cost, or at a nominal fee to offset the cost of creating/hosting the datasets, this will not amount to a commercial gain for the Organizations, and the obligation to pay royalties would not arise. However, Organizations must still ensure that all other considerations due to the voice artists are paid in a timely manner prior to releasing the Speech Datasets in the public domain.

V. Regulatory Liability

Similar to the legal framework around ownership rights, various elements of a Speech Dataset may contain Personal Data and consequently give rise to compliances under India data protection laws such as providing a privacy notice and obtaining prior consent. While the Privacy Rules continue to remain in force until the DPDPA is implemented, it is recommended for Organisations to implement data protection measures that are aligned with the DPDPA to be in a favourable position once it is implemented. This section, therefore, assesses Speech Datasets in the context of applicability and Data Protection Compliances required under both laws.

As a central governing principle, to avoid attracting any compliance requirements under Indian data protection laws, Organisations should aim to avoid the collection of any Personal Data at the very outset, unless absolutely necessary. This may be done for various elements of the Speech Datasets as captured in the points below.

(A) Personal Data contained in underlying text

Where the voice artists are required to recite excerpts of text or otherwise in the form of a script, depending on its nature and source, such text excerpts may contain details about an individual which can be used to identify them. To avoid incorporating such Personal Data as a part of the voice recordings, Organisations can consider:

- conducting prior screening of underlying text to ensure that it does not contain any personal identifiers which may disclose the identity of any other person in the recordings;
- if obtaining text excerpts from third-party sources, contractually ensuring that the text obtained does not contain any personal identifiers.

(B) Personal data of voice artists

Information identifying a voice artist may be collected or created at various stages of the Speech Dataset lifecycle, such as:

(1) Service data

Organisations may collect various categories of information from voice artists when engaging them to provide voice recording services, such as name, contact details, bank account details for payment, etc. Such Personal Data may need to be mandatorily collected for the records of the Organisations, to make payments, and for communications, and Organisations may not be able to avoid such collection. Organisations would therefore need to comply with the Data Protection Requirements, including providing a notice and obtaining prior consent for collection which may be done through the Consent Form.

(2) Metadata

Voice recordings may be required to be complemented with metadata such as age, sex, noise levels, location details, waveforms, disability status, spectrograms and file metadata for each sample, and as a result the metadata may contain Personal Data. However, Organisations can take measures to ensure that the metadata is not capable of being traced back to a specific individual i.e. if the voice artist's name is not collected and the metadata cannot be combined to identify the voice artist. If metadata is created in this manner, it can be argued that the metadata would not amount to Personal Data. Further, in case metadata includes identifiable information, Organisations can explore options to anonymize such data to minimize any potential compliance obligations under Indian data protection laws. Anonymization models must be language-agnostic and must ensure that it incorporates features which limits the Speech Datasets from (a) being linked to a single individual, (b) singling out individuals, and (c) re-identifying an individual based on observed traits. This would also be crucial for ensuring that all processing activities are undertaken in a privacy-preserving manner.

(3) Voice data

There may be instances where voice artists may be required to deliver an extempore or participate in a non-scripted conversation for the purpose of the voice recording. In such cases, there is a possibility that voice artists may inadvertently disclose Personal Data. To avoid such collection, Organisations must undertake measures such as:

- screening the voice recordings and transcripts to ensure that it does not include any personal identifiers and in case any personal identifiers are detected, such portions are redacted; and
- Prohibiting voice artists from disclosing any personal details such as name, address in the recordings that can be traced back to them or any other individuals as identifiers in any way. This may be contractually agreed through the Consent Form.

The voice recording itself may also amount to Personal Data, if, when combined with other information about the voice artist (such as service data and metadata), it is capable of identifying the voice artist. Additionally, the scope of data protection law would extend to the measurement and analysis of voice patterns for authentication purposes. Organisations should, therefore, take measures to ensure that:

- voice recordings are not linked to service data in any way, and further ensure that it does not create metadata of the voice recording in such a way that allows the voice artist to be identifiable.
- voice recordings are not used for voice-printing or to measure/analyse voice patterns for authentication purposes or otherwise obtain any information about the voice artists.

(C) Personal Data of Users

In case Organisations collect any Personal Data (such as email addresses, IP address, cookies data) for the purpose of distributing the Speech Datasets to end-users, Organisations must ensure compliance with the Data Protection Compliances, including ensuring that the website hosting such Speech Datasets requires end-users to consent to the processing of their Personal Data.

VI. Ethical Considerations

In addition to legal considerations, the creation of Speech Datasets necessitates Organisations to adopt measures that minimise ethical concerns that may emerge given its downstream use as fundamental blocks in artificial intelligence models and speech language technologies. Through these efforts, the organization can show the effort that has been placed to minimise risks and prevent harms, thereby acting as a safeguard for saving reputation of the Organisations in case of any unintended consequences arising from its subsequent usage. Organisations engaged in the creation of datasets must strive to ensure that the dataset is created and used in a way that is transparent, representative, and respectful of individual autonomy, confidentiality, and dignity. By proactively addressing ethical concerns, Organisations can contribute to the responsible development and deployment of open-source datasets, fostering trust among users and mitigating potential risks of harm or discrimination.

Various ethical considerations which may become relevant at different stages across the Speech Dataset lifecycle, have been addressed in detail in this section.

(A) Considerations at the collection stage

(1) *Informed and voluntary participation*:

Organisations must ensure that the voice artists have clear information regarding the purposes for using their voice recordings, and these are explained to them in a way they understand. They must have the option to decline participation at any time prior to the signing of the Consent Form without any consequences.

(2) *Representative datasets*:

Organisations must ensure diversity and inclusivity in the dataset to prevent bias and ensure fair representation of different demographics, including age, gender, ethnicity, and language. Organisations must vet the pool of voice artists to avoid over-representation or under-representation of certain minority groups, which could lead to biased algorithms and discriminatory outcomes. Given that regional languages may also have various dialects, Organisations must ensure that they develop datasets that account for these differences.

(3) *Data Quality Assessment*:

Organisations must ensure consistency in the quality of the speech data collected, including by listening to the full audio recording at the time of screening, as it is essential to maintain the integrity and reliability of the dataset for downstream tasks. Organisations must also focus on relying on global standards from the time of collection to ensure interoperability across various speech technologies.

(B) Considerations at the development stage

(1) *Assessing Speech Datasets against potential harms*:

Organisations must assess the Speech Datasets against potential harms that may arise from its downstream use in various speech technologies and adopt measures to mitigate such harms. For instance, Organisations can consider maintaining a checklist of harms which could indicatively include harms arising from bias resulting in unfair or discriminatory outcomes, re-identification of individuals from anonymized or pseudonymized speech data, social or cultural impacts that may exacerbate existing inequalities, and potential misuse of Speech Datasets for unethical or harmful purposes, such as propaganda, misinformation or manipulation of public opinion. Please refer to Annexure-III for an illustrative list of harms.

(2) *Appointing native speakers or trained personnel as reviewers*:

In cases where annotation of the voice recordings takes place through automated means to be later reviewed by human annotators or directly by human annotators, Organisations must consider the socio-linguistic factors that could have an influence on the quality of the annotations. Regional languages may contain certain social and cultural undertones that may only be known to a native speaker or a person with high proficiency. Therefore, to mitigate any errors, Organisations must recruit annotators that have extensive knowledge of dialects or accents appearing in the speech corpus and are familiar with the vocabularies included. The annotators must ensure that they review the entire audio corpus, if feasible.

(C) Considerations at the distribution stage

(1) *Documentation:*

Organisations must ensure that Speech Datasets are accompanied with documentation which includes information on the purpose, creation, composition and distribution of the dataset, as well as providing information on ethical considerations such as diversity, inclusion and privacy protection accounted for at the time of creating the datasets. The documentation must also illustrate the limitations of the datasets and provide metrics which would aid the decision-making process of individuals utilising the datasets.

(2) *Explainability and accountability*:

Organisations should focus on providing clear explanations about the origin of the speech data, how it was collected, and the context of its application to enhance transparency and accountability in the metadata of the voice recordings.

(3) *Feedback Loops*:

Organisations must establish mechanisms to enable review of the Speech Datasets once they are made available in the public domain. Such mechanisms must enable stakeholders to flag issues or concerns in relation to the datasets and Organisations must be able to take appropriate actions to address the concerns. For instance,

Mozilla Common Voice Dataset adopts multiple asynchronous pathways to suggest changes, flag issues and share ideas (through public roadmap, github backlog, discourse, matrix, AMAs, open events). Issues flagged by users must be well documented and must also be accounted for future assessments conducted during the development phase. Such pathways must also be fine-tuned keeping in mind the regional communities that may be interested in actively contributing to the Speech Datasets.

(4) *Review Board*:

Establishing a Review Board can help ensure that ethical standards are upheld throughout the Speech Dataset lifecycle. The Review Board can operate independently of the Organisations creating the Speech Datasets and can comprise of experts of diverse expertise spanning across ethics, linguistics, law, data science, and human rights. The Review Board could serve important functions such as developing clear guidelines and criteria for evaluating Speech Datasets, taking into account relevant ethical principles, legal requirements and industry standards as well as adapting existing frameworks such as the Fairness, Accountability, and Transparency (**FAT**) guidelines or Ethical AI Principles developed by leading research institutes and the scientific community.

(5) *Fees and Charges*:

Organisations should consider the impact of charging fees for downloading Speech Datasets. Even charging nominal fees for download could result in large resource-rich entities getting preferential access to Speech Datatsets, resulting in market disparities, increased network effects, and a slowdown of innovation by smaller players and start-ups.

EXECUTIVE SUMMARY OF RECOMMENDATIONS

Considerations Lifecycle Stage	Legal Considerations (Ownership)	Legal Considerations (Liabilities)	Ethical Considerations
Collection	Voice recordings: Consent Forms with voice artists to contain Assignment Requirements to transfer all IP rights, with adequate compensation. If procured from a third-party, ensure IP rights are transferred in line with Assignment Requirements and other contractual requirements. Text/Transcript: If created by/procured from a third-party, ensure IP rights are transferred in line with Assignment Requirements and other contractual requirements. If collected from open-source databases, comply with applicable license conditions. Adopt quality checks to identify/modify copyrighted content.	Avoid collecting Personal Data unless necessary. If necessary to engage voice artists, obtain their consent through a Consent Form; ensure compliance with Data Protection Obligations. Prohibit voice artists from disclosing Personal Data in recordings. Contractually ensure third-party text does not contain Personal Data.	Ensure informed and voluntary participation of voice artists. Vet voice artist pool to ensure adequate representation of minority groups, languages, age, ethnicity, gender. Account for regional dialects when collecting data.
Curation/Development	Employment/engagement agreements with Personnel to contain confidentiality clauses and Assignment Requirements to transfer all IP rights.	Screen text/voice recordings to detect and remove Personal Data. Ensure metadata cannot be traced back to the voice artists; anonymise metadata. Restrict voice data from being linked to service data or from being analysed for authentication purposes. Recordings not to be used to measure/analyse voice patterns for authentication.	Listen to full audio to ensure consistency in quality and to maintain integrity and reliability of data. Create a checklist of harms to assess Speech Datasets for potential harms in downstream use. Appoint native speakers or trained personnel as reviewers to address social and cultural sensitivities in voice recordings.
Deployment/Sharing	Choose Open Content License best suited to the purpose of sharing.	Comply with Data Protection obligations in case of Personal Data collected from end-users.	Provide comprehensive documentation and clear explanations about datasets and metadata.

	Comply with licensing requirements applicable to any underlying open-source data that may have been used when creating the dataset. Settle all payments to voice artists prior to release. If shared for commercial use, royalty payments may be payable to voice artists.		Establish mechanisms to flag issues or concerns. Establish a Review Board to formulate and uphold ethical standards. Consider the impact of charging fees given risks of preferential access

ANNEXURE I

Data Protection Compliances

1. Information Technology (Reasonable Security Practices and Sensitive Personal Data or Information) Rules, 2011 (Privacy Rules)

Entities that collect, store, process, transfer or disclose PI or SPI of natural persons (**Data Subjects**) are subject to certain obligations under the Privacy Rules:

(i) ***Privacy Policy Requirements:*** The entity would be required to publish a privacy policy containing the information mentioned below:

- clear and easily accessible statements of its practices and policies;
- type of PI or SPI collected;
- purpose of collection and usage of such information;
- disclosure of SPI; and
- reasonable security practices and procedures adopted by the entity.

(ii) ***Data Subject Rights:*** Data Subjects must be given the option to review and amend the PI provided by them.

((i) and (ii) above together the **PI Obligations**)

In addition to the PI Obligations, an entity that collects, stores, processes, discloses, and transfers SPI must comply with certain additional obligations listed below (collectively, **SPI Obligations**):

(iii) ***Consent:*** Obtain the consent of the Data Subject in writing (including by way of ***electronic*** means) prior to the collection, stating the requirements listed below (collectively, **Consent Requirements**):

- The fact that information is being collected;
- the purpose for which the information is being collected;
- the intended recipients of the information;
- the name and address of the agency that is collecting the information; and
- the name and address of the agency that will retain the information.

The Data Subject also has the ability to decline consent prior to collection or withdraw consent at any time by notifying the entity in writing about it. The body corporate will be free to not provide services for which the SPI was sought.

(iv) ***Collection Limitation:*** Collect SPI only for lawful purposes and only if necessary;

(v) ***Data Retention:*** Ensure that SPI collected is not retained for longer than is required for the lawful purpose for which it was collected, or for purpose mandated by law;

(vi) ***Disclosure:*** Obtain permission of the Data Subject before disclosing such information to any third party, unless such disclosure has been contractually agreed to between the parties, or is required for the compliance of a legal obligation;

(vii) ***Transfer:*** Before transferring such information to any other body corporate or person in India or overseas, ensure that the transferee has implemented the same level of data protection that is prescribed under the Privacy Rules and adhered to by the transferring entity, and that such transfer is pursuant to a lawful contract or after obtaining consent from the Data Subject;

(viii) ***Data Security:*** Implement security practices and standards and have a comprehensive documented information security programme and information security policies in place. The IS/ISO/IEC 27001 is one such standard specified in the Privacy Rules that bodies corporate may implement. Additionally, standards notified by industry associations and approved and notified by the Central Government may also be implemented. If either the ISO standard or an industry association standard is implemented, it must be certified or audited on a regular basis, and at least once a year or as and when the entity undertakes a significant upgradation of its process and computer resource through an independent auditor, approved by the Central Government; and

(ix) ***Appointment of Grievance Officer:*** Appoint an officer to redress the grievances of Data Subjects with respect to processing of information, within one month from the date of receipt of such grievance (**Grievance Officer)**. For this purpose, the body corporate must publish the Grievance Officer's name and contact details on its website.

Pursuant to Press Note dated 24 August 2011, the Ministry of Electronics and Information Technology (**MeitY**) has clarified that an entity which, on behalf of another entity, collects, stores, processes, transfers or discloses PI or SPI of a Data Subject under a contractual obligation is required to ensure compliance with requirements (a), (g) and (h) mentioned above (**Data Processor Obligations**).

The violation of any of the obligations under the Privacy Rules would attract a fine up to INR 1,00,000. Furthermore, if a provider of information suffers wrongful loss due to the body corporate's negligence in securing the information by implementing reasonable security practices, then the body corporate would also be liable to compensate such an individual.

2. Digital Personal Data Protection Act, 2023

(i) ***Personal Data:*** As per the DPDP Act, Personal Data is defined to mean "*any data about an individual who is identifiable by or in relation to such data.*" The DPDP Act, once enforced, will create a uniform compliance structure for all types of personal data, irrespective of their sensitivity. In this regard, the DPDP Act applies to the processing of all digital Personal Data: (a) within India, irrespective of whether the Data Principal is based in India or not; or (b) outside India in relation to offering goods or services to Data Principals in India, by Data Fiduciaries or by Data Processors.

The DPDP Act specifically does not apply to (i) non-automated processing of Personal Data; (ii) Personal Data processed for personal or domestic purpose and (iii) if the Personal Data is made publicly available.

The DPDP Act also provides for certain exemptions. As relevant in the present case, none of the provisions under the DPDP Act would apply in respect of processing of Personal Data necessary for research, archiving or statistical purposes, if the Personal Data is not used to take any decision specific to a Data Principal and such processing is carried on in accordance with such standards as may be prescribed.

(ii) ***Data Fiduciaries and Data Processors:*** The DPDP Act imposes various obligations (outlined in further detail below) on Data Fiduciaries, i.e., entities that determine the purpose and the means by which the Personal Data of Data Principals i.e., individuals to whom the data relates, must be processed. To carry out such processing, a Data Fiduciary may engage, appoint, use, or otherwise engage third parties i.e., Data Processors only under a valid contractual arrangement.

The DPDP Act imposes the primary obligation of compliance on the Data Fiduciary including in cases where the Personal Data is processed by a Data Processor on its behalf. The Data Fiduciary is required to *inter alia*: (i) protect Personal Data in its possession or under its control, including in respect of any processing undertaken by it or on its behalf by a Data Processor, by taking reasonable security safeguards to prevent Personal Data breach; (ii) erase Personal Data upon the Data Principal withdrawing her consent soon as it is reasonable to assume that the specified purpose is no longer being served, whichever is earlier unless necessary for compliance with applicable law and accordingly cause the Data Processor to erase such data.

(iii) ***Grounds to Process Personal Data:*** Data Fiduciaries are permitted to process (including collecting, storing, using, disclosing etc.) Personal Data only for lawful purposes, either based on the Data Principal's consent or for certain identified legitimate use cases. These grounds are explained in further detail below:

(a) *Consent*

Personal Data may be processed if the Data Principal provides consent to such processing. Consent for processing Personal Data must be free, specific, informed, unconditional, and capable of being withdrawn, in addition to being demonstrated through clear, unambiguous, affirmative action for a specified purpose and be limited to such Personal Data that is necessary for the specified purpose.

Notice Requirements: The Data Fiduciary would be required to provide notice to Data Principals at the time of requesting consent or as soon as reasonably practicable for the processing of Personal Data conducted on the basis of past consent.

The notice must inform the Data Principal of: (i) the Personal Data and the purpose for which the same is proposed to be processed; (ii) the manner in which the Data Principal may (a) exercise their rights of withdrawal; (b) grievance redressal; and (c) make a complaint to the Board. However, the Central Government may prescribe the format of such notice later.

(b) *Legitimate use*

The DPDP Act recognizes certain circumstances in which express consent is not required and processing of Personal Data is permitted for limited instances such as (i) compliance with a judgment or order; or (ii) processing in public interest (e.g., processing in the interest of sovereignty and integrity of India and security of the State, fulfilling any obligation under any law, provision of subsidies, benefits etc.)

(iv) ***Processing of Personal Data of children/persons with disabilities***

The DPDP Act imposes additional obligations for processing children's Personal Data. A child means any individual under the age of 18 years. The DPDP Act requires verifiable parental consent (i.e., consent of the parental or lawful guardian) to be obtained in the manner subsequently prescribed by the Central Government in relation to children or persons with disabilities. The Data Fiduciary is also restricted from tracking, monitoring behaviour of, or targeting advertisements directed at children, or undertaking any other processing that is likely to cause harm to children (as may be prescribed by regulations). The Central Government by notification may exempt or partially exempt entities in relation to the restrictions on the processing of children's data, subject to certain conditions as may be prescribed.

(v) ***Data Fiduciary Obligations:*** Data Fiduciaries that are either directly engaged in processing or through Data Processors are subject to certain obligations under the DPDP Act:

(a) *Erasure of Personal Data*

The Data Fiduciary, unless retention is necessary for compliance with any law, is required to erase Personal Data when a Data Principal withdraws her consent, or as soon as it is reasonable to assume that the specified purpose for which the Personal Data was collected is no longer served (whichever is earlier). The Central Government is yet to notify the time period within which it is deemed that the specified purpose is no longer served.

(b) *Completeness, accuracy, and consistency*

The Data Fiduciary must ensure the completeness, accuracy and consistency of the Personal Data processed, if such Personal Data is used to make a decision that affects a Data Principal or is disclosed to another Data Fiduciary.

(c) Notification of Personal Data breaches

The Data Fiduciary must intimate the Board and each affected Data Principal of every Personal Data breach. Personal Data breach is defined to mean any unauthorized processing of Personal Data or accidental disclosure, acquisition, sharing, use, alteration, destruction, or loss of access to Personal Data, that compromises the confidentiality, integrity, or availability of Personal Data.

Upon receiving intimation of a breach from a Data Fiduciary, the Board may direct the Data Fiduciary to undertake urgent remedial or mitigation measures to minimise the impact of the breach. Additionally, the Board may investigate a Personal Data breach and levy penalties based on either an intimation received from the Data Fiduciary or a complaint lodged by the Data Principal.

(d) Technical safeguards and reasonable security measures

Data Fiduciaries must implement appropriate technical and organizational measures to ensure that they effectively observe the provisions of the DPDP Act and take reasonable security measures to prevent Personal Data breaches.

(e) Grievance Redressal Mechanism

Data Fiduciaries must publish the business contact information of a person who may answer a Data Principal's queries pertaining to the processing of their Personal Data and the Data Fiduciary must establish an effective grievance redressal mechanism.

(vi) Rights of Data Principals

Enable Data Principals to exercise their right to request and obtain the following information from Data Fiduciaries to whom they have given consent: (a) a summary of Personal Data being processed, and processing activities in this respect; (b) identities of all Data Fiduciaries and Data Processors with whom the data has been shared; and (c) any other information related to the Personal Data or its processing as may be prescribed. Save for limited exceptions (such as, where data has been shared with another Data Fiduciary for the investigation of offences), there is no scope to refuse requests of this nature.

Data Principals also have the right to request the Data Fiduciary to correct, complete, or update their Personal Data and seek the erasure of their Personal Data in a form that may be prescribed by the Central Government. Data Fiduciaries must act on such requests to erase data unless this data is necessary for: (a) the specified purpose; or (b) for compliance with laws in force.

Additionally, Data Principals have the right to nominate any other individual who in the event of the death or incapacity of the Data Principal will be responsible for exercising their rights in accordance with the provisions of the DPDP Act, and the right to have their grievances redressed.

(vii) Cross-border transfer of Personal Data

Data Fiduciaries are not permitted to transfer Personal Data to countries that the Central Government may blacklist for this purpose (such a list has not yet been prescribed). Additionally, the Central Government is also empowered to prescribe conditions for transfer generally.

(viii) Penalties

The DPDP Act introduces specific financial penalties for significant contraventions of its provisions, that may be up to INR 250 crore, depending on the nature of the contravention.

ANNEXURE – II

Open-Source Licenses: Overview

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ANNEXURE – III

Illustrative List of Harms

1. ***Privacy and Consent Violations***: In case Speech Datasets contain Personal Data of voice artists such as names, addresses, or other contextual information, such information may be used to identify individuals. Therefore, unconsented use of such Personal Data by entities would lead to a breach of their privacy rights.

   Checklist: Take steps such as screening the underlying text, transcripts, and voice recordings to ensure that Personal Data does not form a part of the Speech Datasets. Additionally, to the extent the Speech Datasets may contain Personal Data, ensure compliance with the Data Protection Compliances. Further, ensure that voice artists have clear information regarding the purposes for using their voice recordings and these are explained to them in a way they understand.

2. ***Risk of re-identification:*** Even anonymised information of voice artists may be at the risk of re-identification through techniques such as mosaic theory of intelligence gathering in which disparate pieces of information become significant when combined with other types of information. Even if data is anonymized, with efforts made to remove personal identifiers, if there are multiple datasets containing similar or complementary information, it may be possible to determine identity based on various data points, such as gender, location, educational status, etc. across such datasets.

   Checklist: Consider deploying privacy specific technological measures in the interest of protection data such as differential privacy or homomorphic encryption and adopt mechanisms/techniques to prevent re-identification from anonymised data.

3. ***Misuse:*** Speech Datasets could be used for unintended or unauthorized purposes, such as creating fake identities or "deepfakes". This could lead to risks such as misinformation, fraud, or impersonation.

   Checklist: Minimise Personal Data collection through measures such as screening the underlying text, transcripts and voice recordings to remove Personal Data from Speech Datasets ensuring that even in the event Speech Datasets are used for creation of deepfakes, it cannot be attributed to a particular individual.

4. ***Bias and Discrimination***: Models trained on under-representative Speech Datasets which do not accommodate diverse accents, dialects, or demographics could potentially amplify the risk of bias and discrimination in downstream use-cases. Datasets that are not inclusive may result in the development of technologies that exclude or disadvantage certain groups, such as those with speech impairments and non-native speakers.

   Checklist: At the collection stage, adopt measures such as ensuring diversity and inclusivity in the dataset to prevent bias and ensure fair representation of different demographics, including age, gender, ethnicity, and appoint native speakers or trained personnel as reviewers to prevent mitigate any errors at the development stage.

5. ***Addressing Profanities:*** Profanities are subjective and culturally dependent. What is considered as offensive in one culture might be commonplace in another. Therefore, in the event annotators are not provided clear guidelines on what constitutes profanity, and how to handle it, or due to lack of monitoring, inconsistencies can arise leading to unintended inclusion of offensive language.

6. ***Checklist:*** Assess Speech Datasets against potential harms arising from profanities at the development stage, and establish mechanisms to review the Speech Datasets through feedback loops at the distribution stage and ensure that ethical standards determined by the Review Board are upheld throughout the Speech Dataset Lifecycle. Further, consider engaging native speakers or trained personnel as reviewers to mitigate the risks of cultural profanities.***Inaccuracy and Misrepresentation***: Incomplete or low-quality Speech Datasets used as training data could result in models that inaccurately or inadequately transcribe or understand speech, affecting user experience and potentially leading to dangerous misinterpretations in critical applications.

   Checklist: Ensure consistency in the quality of the speech data collected, including by listening to full audio recording at the time of screening, to maintain the integrity and reliability of the dataset for downstream tasks.

7. ***Model Drift and Decay***: Where models are trained on Speech Datasets that are not properly managed, scaled or maintained, such models might not adapt well to evolving language patterns, leading to decreased accuracy over time. This drift could impact voice-activated systems requiring continuous updates.

   Checklist: Maintain documentation along with the Speech Datasets which provides details on the status of the dataset, and establish feedback loops to enable review of the Speech Datasets once they are made available in the public domain.